Midterm Leftover

Inverse Function Theorem

If $f(\mathbf{a}) = \mathbf{b}$ and the derivative matrix $[Df(\mathbf{a})]$ is invertible ($\det \neq 0$), then locally f has a unique inverse function.

- **The Core Formula:** $[D(f^{-1})] = [Df]^{-1}$.
- **The Trap:** If $\det([Df]) = 0$, the theorem fails to guarantee an inverse, but one might still exist depending on higher derivatives.

Implicit Function Theorem

Allows finding derivatives without solving for the explicit function.

- **Planar Curves ($f(x, y) = k$):** Beware the minus sign!
$$\frac{dy}{dx} = -\frac{D_x f}{D_y f}$$
- **The Product Trap:** Partial derivatives are NOT quotients.
$$\left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial x}{\partial z}\right)\left(\frac{\partial z}{\partial y}\right) = -1$$
- **Spatial Curves ($f = k_1, g = k_2$):** The curve is the intersection of two surfaces. The tangent direction is simply $\nabla f \times \nabla g$.
- **General Matrix Form:** If separating into free variables (x) and dependent variables (y): $[Dy] = -[D_y f]^{-1}[D_x f]$.
- **The Sharp Turn Flag:** If $\nabla f = \mathbf{0}$, the theorem fails. This usually signals a sharp corner or a crossing where no unique tangent exists.

Vector Field Derivatives

- **Fixed Points ($F(p) = 0$):** Analyze the eigenvalues of $[DF(p)]$.
  - *Source:* All eigenvalues have positive real parts (repelling).
  - *Sink:* All eigenvalues have negative real parts (attracting).
  - *Saddle:* Mixed positive and negative real parts.
- **Divergence ($\nabla \cdot \mathbf{F}$):** Measures flux/spawning.
$$\nabla \cdot \mathbf{F} = \text{trace}([D\mathbf{F}]) = D_1F_1 + D_2F_2 + D_3F_3$$
If > 0 it is a source (faucet). If < 0 it is a sink (drain). If $= 0$ it is incompressible/solenoidal.
- **Curl ($\nabla \times \mathbf{F}$):** Measures rotation axis and speed.
$$\nabla \times \mathbf{F} = \langle D_2F_3 - D_3F_2,\ D_3F_1 - D_1F_3,\ D_1F_2 - D_2F_1 \rangle$$
For 2D fields, it is just the scalar $D_1F_2 - D_2F_1$. If $= 0$, it is spinless.
- **Laplacian (Δf):** The divergence of the gradient field. $\Delta f = \nabla \cdot (\nabla f)$.

The Great Vector Identities

These are the multivariable cheat codes born from Clairaut's Theorem:

- **Gradients are Spinless:** $\nabla \times (\nabla f) = \mathbf{0}$
- **Curls are Solenoidal:** $\nabla \cdot (\nabla \times \mathbf{F}) = 0$
- **Leibniz (Product) Rules:**
  - $\nabla \cdot (f\mathbf{F}) = (\nabla f) \cdot \mathbf{F} + f(\nabla \cdot \mathbf{F})$
  - $\nabla \times (f\mathbf{F}) = (\nabla f) \times \mathbf{F} + f(\nabla \times \mathbf{F})$

Integration & Fubini

Finding Bounds (The Two Ways)

- **Geometric Way:** Draw the 2D shadow. Shoot an arrow in the direction of the inner integral (e.g., UP for dy). Where the arrow *enters* the region is your lower bound function; where it *exits* is your upper bound function.
- **Dumb Computer Way:** Take the boundary equations and strictly solve them for the inner variable (e.g., $y = x^2$). These are your inner bounds. Find the absolute numerical min/max intersections for your outer bounds. Outer bounds MUST be constants.

Fubini's Order-Swapping Hack

If the inner integral is literally impossible (e.g., $\int e^{x^2}\,dx$ or $\int \sin(y^2)\,dy$), **swap the order of integration**!

1. Sketch the 2D shadow using the current bounds.
2. Swap horizontal slicing ($dx\,dy$) for vertical slicing ($dy\,dx$).
3. Find the new bounds using the Geometric Way.

Coordinate Transformations

Used for slanted regions ($x + y = 1$) or ellipses. You MUST include the Jacobian multiplier $|J|$. Using variables (u, v):
$$dx\,dy = \left|\det \begin{vmatrix} D_u x & D_v x \\ D_u y & D_v y \end{vmatrix}\right| du\,dv$$
The Inverse Hack: If your custom equations are given as $u(x, y)$ and $v(x, y)$, it is faster to compute the inverse Jacobian: $J^{-1} = (D_x u)(D_y v) - (D_y u)(D_x v)$. The multiplier you need is simply $|J| = 1/|J^{-1}|$.

The Standard Coordinate Systems

Used when integrands/bounds feature $x^2 + y^2$ or $x^2 + y^2 + z^2$.

- **Polar (2D) (r, θ):** Best for flat circles/arcs.
$x = r\cos\theta, \quad y = r\sin\theta.$
Multiplier: $dA = \mathbf{r}\,dr\,d\theta$
- **Cylindrical (3D) (r, θ, z):** Best for pipes/disks/paraboloids.
$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$
Multiplier: $dV = \mathbf{r}\,dr\,d\theta\,dz$
- **Spherical (3D) (ρ, ϕ, θ):** Best for domes/spheres.
$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi.$
ϕ is the angle dropping from the top z-axis ($0 \le \phi \le \pi$).
Multiplier: $dV = \boldsymbol{\rho^2}\sin\phi\,d\rho\,d\phi\,d\theta$

Surface Integrals (Scalar Fields)

The Core Formula

Integrating a physical density ρ over a 3D surface S requires a parametrization map $\Phi(u, v)$.
$$\int_S \rho\,dS = \int_U \rho(\Phi(u, v))||D_u\Phi \times D_v\Phi||\,dA_{u,v}$$
The vector length $||D_u\Phi \times D_v\Phi||$ is your "Area Scaling Factor" (the surface Jacobian).

The "Lazy" Hacks for dS

Do NOT compute horrific cross-products if you don't have to. Memorize these three instant shortcuts:

- **Graph of a Function ($z = f(x, y)$):** If the surface is given directly as a function over a 2D shadow, you bypass parametrization entirely. $dS = \sqrt{1 + ||\nabla f||^2}\,dA_{x,y}$.
- **Spherical Shell:** For a sphere of constant radius R, the multiplier is identical to 3D spherical coordinates, just without the dr. $dS = R^2\sin\phi\,d\phi\,d\theta$.
- **3D Pythagorean Theorem:** If your region U is just a flat shape sitting on a tilted plane, its true area is locked to its 2D shadows on the coordinate planes.
$$\text{Area}(U)^2 = \text{Area}(U_{xy})^2 + \text{Area}(U_{xz})^2 + \text{Area}(U_{yz})^2$$

Applications (Physics & Prob)

Probability vs. Center of Mass

Physics and Probability use the EXACT same formulas. In probability, the "total mass" is just total probability, which is always 1, so the denominator completely vanishes!

- **Total Mass (m):** $m = \int_D \rho\,dV$. (In Probability, $m = 1$).
- **Moment of Mass (M_x, M_y):** The "effect" of mass pushing down on a lever. Multiply the density by the distance to the axis of rotation. $M_x = \int_D x\rho\,dV$.
- **Center of Mass / Expected Location:** Divide the moment by total mass. $\overline{x} = M_x/m$.
- **Expected Value:** $\mathbb{E}(f(X)) = \int f(x)\rho(x)\,dV$.

Moment of Inertia (I) / Angular Mass

Measures how stubbornly an object resists being spun around an axis. Unlike regular mass, distance is SQUARED.
$$I_L = \int_D (\text{distance to axis } L)^2 \rho\,dV$$

- **Around z-axis:** Distance squared is $x^2 + y^2$. $I_z = \int_D (x^2 + y^2)\rho\,dV$.
- **Around x-axis:** Distance squared is $y^2 + z^2$. $I_x = \int_D (y^2 + z^2)\rho\,dV$.

Topology & Boundaries

The Golden Rule

Identify the shape before doing any math!

- **Closed Surface (Balloon):** Encloses a 3D volume. No edges. Ant walks forever. ($\partial S = 0$) $\to$ **Divergence Thm**.
- **Open Surface (Bowl):** 2D fabric with a 1D rigid boundary wire. Ant falls off. ($\partial S \neq 0$) $\to$ **Stokes' Thm** or Cap it.
- **The Boundary (∂S):** The 1D rim itself. Must form a loop.

2D Cheat Codes (Green's)

Bypass terrible 2D line integrals around loops by taking a double integral over the flat shadow D inside. Let $\mathbf{F} = \langle F_1, F_2 \rangle$.

Work / Circulation Form (2D Curl)

Measures how badly the field fails the exactness test.
$$\oint_C F_1\,dx + F_2\,dy = \iint_D (D_1F_2 - D_2F_1)\,dA$$

Flux Form (2D Divergence)

Measures 2D spawning/draining (outward flow).
$$\oint_C F_1\,dy - F_2\,dx = \iint_D (D_1F_1 + D_2F_2)\,dA$$

3D Cheat Codes

Stokes' Theorem (Curl / Soap Film)

Translates microscopic spin into macroscopic edge work.
$$\int_{\partial S} \mathbf{F} \cdot d\gamma = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS$$

- **Triggers:** Work along a 3D loop, or Flux of a Curl.
- **Hack (Surface Swap):** The "soap film" shape doesn't matter, only the wire! Swap ugly domes for **flat disks**.
- **Instant Zero:** Flux of a curl over a completely closed surface (no boundary wire) is always 0.

Divergence Theorem (Sealed Box)

Translates outward flux into net spawning/draining inside.

$$\iint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

- **Triggers:** Flux through a completely **closed** 3D shape.
- **Hack (Capping Trick):** If surface is open (e.g., open pipe):
  1. Add flat lids to seal holes (normals point strictly OUT).
  2. Calculate easy 3D Volume Divergence of the closed shape.
  3. Calculate simple flux of the flat lids ($\mathbf{F} \cdot \mathbf{n}_{\text{out}}$).
  4. **Subtract** the lid flux from the Volume Divergence.

Finding Normal Vectors (n)

The Surface Gradient Hack ($z = f(x, y)$)

Never compute the ugly square root! The unnormalized vector $\mathbf{n} \, dS$ automatically absorbs the Area Scaling Factor. Just take two derivatives and you are ready to dot product.

- **Pointing UP (Positive z):** $\mathbf{n} \, dS = \langle -D_x f, -D_y f, 1 \rangle \, dA_{x,y}$
- **Pointing DOWN (Negative z):** $\mathbf{n} \, dS = \langle D_x f, D_y f, -1 \rangle \, dA_{x,y}$

Parametrized Surfaces ($\Phi(u, v)$)

If you must parametrize, the normal vector combined with dS is just the cross product:

$$\mathbf{n} \, dS = (D_u \Phi \times D_v \Phi) \, dA_{u,v}$$

Trap Check: Look at the z-component of your result. If the problem asks for "upward" and your z-component is negative, just multiply the entire vector by -1.

Stokes' Theorem (Right-Hand Rule)

The orientation of your "soap film" ($\mathbf{n}$) MUST match the direction the ant walks on the boundary wire.

- Loop is **Counter-Clockwise (CCW)** from above $\rightarrow$ $\mathbf{n}$ points **UP**.
- Loop is **Clockwise (CW)** from above $\rightarrow$ $\mathbf{n}$ points **DOWN**.

Closed Surfaces (Divergence / Capping)

Normals must point strictly **OUTWARD** (away from the trapped 3D volume).

- **Top Flat Lid** ($z =$ constant): $\mathbf{n} = \langle 0, 0, 1 \rangle$
- **Bottom Flat Lid** ($z =$ constant): $\mathbf{n} = \langle 0, 0, -1 \rangle$
- **Vertical Cylinder Wall** ($x^2 + y^2 = R^2$): $\mathbf{n} = \langle x, y, 0 \rangle / R$

Infinite Series (Ch. 16)

Absolute vs. Conditional

- **Absolute:** $\sum |a_n|$ converges. You can rearrange terms legally.
- **Conditional:** Only converges due to alternating signs. **TRAP:** Rearranging terms changes the final sum (Associativity/Commutativity rules fail).

Absolute Convergence Decision Tree

1. **Limit $\neq 0$?** Diverges. Stop calculating.
2. **Looks integrable?** Integral test!
3. **Has $n!$ or c^n? Ratio Test** ($\lim |\frac{a_{n+1}}{a_n}| < 1$).
4. **Has $[f(n)]^n$? Root Test** ($\lim |a_n|^{1/n} < 1$).
5. **Ratio/Root = 1?** Useless. Use **Limit Comparison (LCT)**. Compare highest powers to a p-series ($1/n^p$).

Conditional Convergence (The Balancers)

- **Leibniz Test:** Strict alternating $(+, -, +, -)$. If terms shrink to exactly 0, it conditionally converges.
- **Dirichlet Test:** (For chaotic signs like $\sin(n)$). Split into $A_n \times B_n$. If A_n is a bounded wave (never escapes a ceiling) AND B_n shrinks steadily to 0, it converges.
- **Exam Trap:** $\sin(n)$ and $\cos(n)$ partial sums are strictly bounded. State "given fact from lecture notes" and proceed.

Power Series (Ch. 17)

Interval of Convergence (The Blast Radius)

A polynomial disguise $\sum a_n (x - x_0)^n$ only works inside the Radius R.

1. **Find R (Bulletproof Root Test):** Use $R = \frac{1}{\limsup |a_n|^{1/n}}$.
   - Limit $= 0 \rightarrow R = \infty$ (Converges everywhere)
   - Limit $= \infty \rightarrow R = 0$ (Converges nowhere)
2. **Find R (Practical Ratio Test):** Set $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$. Factor out the $|x - x_0|$ and divide the limit over to get $|x - x_0| < R$.
3. **The Boundary Trap:** You MUST manually plug the exact boundary numbers $(x_0 + R, x_0 - R)$ back into the original series and use Ch 16 rules to see if edges are included $[,]$ or excluded $(,)$.

Note: Doing calculus (derivatives/integrals) on the series term-by-term is legal inside the radius, but it can break the boundary points!

Fourier Analysis (Ch. 18)

The Complex Hack (Frequency Domain)

Never integrate sines/cosines if you can avoid it. Use Euler's: $e^{inx} = \cos(nx) + i \sin(nx)$.

$$\hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} \, dx$$

Extracting Real Coefficients: $a_n = \hat{f}_n + \hat{f}_{-n}$ and $b_n = i(\hat{f}_n - \hat{f}_{-n})$

Fourier Shortcuts

- $\int_{-\pi}^{\pi} x e^{-inx} \, dx = \frac{2\pi i}{n} (-1)^n$; $\int_{-\pi}^{\pi} x^2 e^{-inx} \, dx = \frac{4\pi}{n^2} (-1)^n$
- $\int_0^{2\pi} x e^{-inx} \, dx = \frac{2\pi i}{n}$; $\int_0^{2\pi} x^2 e^{-inx} \, dx = \frac{4\pi}{n^2} + \frac{4\pi^2 i}{n}$

The Symmetry Hacks (Kill Integrals)

- **Even Function (y-axis mirror):** Sines cannot exist. $b_n = 0$.
- **Odd Function (origin spin):** Cosines and DC offset cannot exist. $a_n = 0, a_0 = 0$.

Convergence (Corners vs. Jumps)

- **Corners (e.g., $|x|$):** Function is connected but sharp. Series converges **Uniformly** everywhere.
- **Jumps (e.g., Square Wave):** Function teleports. Series converges Pointwise. **TRAP:** At the exact x-coordinate of the jump, the series evaluates to the **exact average** of the top and bottom cliffs: $\frac{1}{2}(f(x^+) + f(x^-))$.

Parseval's Identity (The God-Tier Hack)

Used to solve impossible infinite sums (like $\sum \frac{1}{n^4}$) by squaring known coefficients.

$$||f||_2^2 = \sum_{n=-\infty}^{\infty} |\hat{f}_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(x)|^2 \, dx$$

Standard Formula Booklet

Standard Integrals

$\int x^n \, dx = \frac{x^{n+1}}{n+1}$; $\int \frac{1}{x} \, dx = \ln |x|$

$\int e^{ax} \, dx = \frac{1}{a} e^{ax}$; $\int a^x \, dx = \frac{a^x}{\ln a}$

$\int \sin x \, dx = -\cos x$; $\int \cos x \, dx = \sin x$

$\int \tan x \, dx = \ln |\sec x|$; $\int \cot x \, dx = \ln |\sin x|$

$\int \sec x \, dx = \ln |\sec x + \tan x|$; $\int \csc x \, dx = \ln |\tan \frac{x}{2}|$

$\int \cos^2 x \, dx = \frac{x}{2} + \frac{\sin 2x}{4}$; $\int \sec^2 x \, dx = \tan x$

$\int \sec^3 x \, dx = \frac{1}{2}(\sec x \tan x + \ln |\sec x + \tan x|)$

$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(\frac{x}{a})$; $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan(\frac{x}{a})$

$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \text{arcsec} \frac{|x|}{a}$; $\int \sinh x \, dx = \cosh x$

Log Forms (Inverse Hyperbolic)

$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2})$ (arsinh)

$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln |x + \sqrt{x^2 - a^2}|$ (arcosh)

$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln |\frac{a + x}{a - x}|$ (artanh)

Power Reduction Identities

- $\sin^2 u = \frac{1 - \cos(2u)}{2}$
- $\cos^2 u = \frac{1 + \cos(2u)}{2}$
- $\tan^2 u = \frac{1 - \cos(2u)}{1 + \cos(2u)}$

Product-to-Sum Identities

- $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$
- $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$
- $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$

Common Taylor Series

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$ ($|x| < 1$)

$\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ ($|x| < 1$)